# Inherited Prekallikrein Deficiency (Fletcher Factor Deficiency): A Comprehensive Disease Characterization

**Disease:** Inherited Prekallikrein Deficiency **MONDO ID:** MONDO:0012901 | **OMIM:** 612423 | **Causal gene:** *KLKB1* (plasma kallikrein B1 / prekallikrein; HGNC:6371; NCBI Gene 3818; OMIM *229000; chromosome 4q35) Synonyms:** Fletcher factor deficiency; congenital/hereditary/severe plasma prekallikrein (PK) deficiency

## Summary

Inherited prekallikrein deficiency is a rare **autosomal-recessive** disorder of the plasma **contact system**, caused by **biallelic loss-of-function variants in *KLKB1***, the gene that encodes the liver-synthesized zymogen **plasma prekallikrein** (also historically called **Fletcher factor**). Prekallikrein circulates bound to high-molecular-weight kininogen (HK) and, once converted to active plasma kallikrein by activated factor XII (FXIIa), participates in two intertwined cascades: it **reciprocally activates FXII** (amplifying intrinsic contact-phase coagulation) and it **liberates bradykinin** from HK (driving vascular permeability and vasodilation via the B2 receptor).

The defining clinical paradox of this disease is that, despite producing a **markedly prolonged activated partial thromboplastin time (aPTT)** in vitro, it causes **no bleeding diathesis**. This is because FXII can slowly auto-activate on anionic surfaces independently of kallikrein, preserving downstream thrombin generation and hemostasis in vivo. The condition is therefore best understood as a **benign laboratory phenotype rather than a true coagulopathy** — typically discovered incidentally during preoperative or routine coagulation screening in an asymptomatic person. The hallmark laboratory signature is an isolated, severely prolonged aPTT with normal PT that **shortens upon prolonged pre-incubation** of plasma with the contact activator, corrects on mixing with normal plasma but **not** with PK-deficient plasma, and is accompanied by **normal thromboelastography (TEG)**.

Prognosis is excellent: a 111-case meta-analysis found only 4 major bleeding events among 96 individuals (prevalence 4%, annualized rate 0.1% — comparable to the general population). The deficiency does **not** protect against thrombosis, and cardiovascular events accumulate with age. No specific treatment is required; fresh frozen plasma (FFP) is used only to normalize the activated clotting time so that heparin anticoagulation can be monitored during surgery. The disease's underlying biology is clinically consequential far beyond the deficiency itself: plasma kallikrein is a validated drug target in **bradykinin-mediated hereditary angioedema (HAE)**, and agents that lower prekallikrein (e.g., the antisense oligonucleotide **donidalorsen**) or inhibit kallikrein (**lanadelumab, berotralstat, sebetralstat**) pharmacologically **phenocopy the deficiency** while proving safe and effective.

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## Key Findings

### 1. The disease is an autosomal-recessive contact-system defect causing isolated prolonged aPTT without bleeding

Severe plasma prekallikrein deficiency is inherited as an autosomal-recessive trait and is characterized by **isolated aPTT prolongation**. The largest synthesis to date (Barco et al. 2020) assembled **111 cases from 89 families** and found only **4 major bleeding events among 96 individuals (3 of them provoked), for a prevalence of 4% and an annualized rate of 0.1%** — indistinguishable from the general population.

> *"Severe plasma prekallikrein (PK) deficiency is an autosomal-recessive defect characterized by isolated activated partial thromboplastin time prolongation."* — PMID: 32202057

The mechanistic reason bleeding is absent is that **FXII retains the ability to auto-activate**:

> *"Because patients with prekallikrein deficiency have preserved auto-activation of factor XII, they rarely manifest with increased bleeding risk or thrombotic risk."* — PMID: 40125216

## 2. Prevalence is low overall but markedly enriched in African ancestry

Using gnomAD allele frequencies and the population-based Gutenberg Health Study, severe PK deficiency prevalence was estimated at **1/155,668 overall and 1/4,725 among individuals of African ancestry** — a ~33-fold enrichment suggesting common/founder loss-of-function *KLKB1* alleles in that population. The ~111 reported cases from 89 families indicate substantial under-ascertainment, consistent with an asymptomatic disorder.

> *"estimated a prevalence of severe PK deficiency of 1/155 668 overall and 1/4725 among Africans"* — PMID: 32202057

## 3. Prekallikrein is a multidomain contact-system zymogen with a dual role in coagulation and inflammation

*KLKB1* encodes plasma prekallikrein, a liver-synthesized zymogen that circulates bound to HK. Structurally it comprises **four apple domains plus one catalytic serine-protease domain**. Factor XI (FXI), a homodimer with the same domain architecture, arose evolutionarily through a duplication of *KLKB1*. Functionally, FXIIa converts prekallikrein to kallikrein, which then reciprocally activates FXII and cleaves HK to release bradykinin, and also activates pro-urokinase (linking the system to fibrinolysis).

> *"it converts plasma prekallikrein to kallikrein (PK), which reciprocally activates FXII and liberates bradykinin from high molecular weight kininogen"* — PMID: 30118759
>
> *"the coagulation factor XI (FXI) arose through a duplication of the gene KLKB1, which encodes plasma prekallikrein"* — PMID: 37874916

## 4. Diagnosis rests on a characteristic aPTT pattern confirmed by mixing studies and specific assay

Severe PK deficiency produces a markedly prolonged aPTT (reported examples: **87.4 s** [reference 24.5–34.0] and **146 s**) with normal PT, thrombin time, and anti-FXa. The aPTT characteristically **shortens with prolonged pre-incubation** of plasma with contact activator — a diagnostic clue — **corrects on mixing with normal plasma but not with PK-deficient**

**plasma**, and is confirmed by a low specific PK assay. Western immunoblot may show absent PK antigen (cross-reactive-material–negative). Critically, **viscoelastic testing (TEG) is normal**, confirming there is no true in vivo coagulopathy.

> *"The aPTT corrected on mixing with normal pooled plasma but not with PK-deficient plasma."* — PMID: 42454615
>
> *"Despite the prolonged aPTT, TEG 6s parameters were normal, including the citrated kaolin R time"* — PMID: 42454615
>
> *"His PTT characteristically became shorter with prolonged incubation, providing a clue at testing for PK levels"* — PMID: 32617222

## 5. Prekallikrein-null mice are hemostatically normal but protected from thromboinflammation

*Klkb1*-knockout (*Klkb1*−/−) mice are viable and fertile with **no spontaneous bleeding**, mirroring the human phenotype. However, they are **protected across multiple thromboinflammatory models**: reduced neurologic deterioration and brain edema in experimental cerebral malaria; attenuated imiquimod-induced psoriatic skin inflammation with abolished bradykinin rise; and protection from extracellular-vesicle–induced pulmonary thrombosis (a protection shared by FXII, HK, and prekallikrein deficiency). Prekallikrein ablation also stabilizes the LDL receptor, lowers LDL cholesterol, and protects against atherosclerosis — revealing unexpected metabolic roles.

> *"Prekallikrein-null (Klkb1-/-), prolylcarboxypeptidase hypomorph ... mice with ECM had reduced neurologic deterioration and brain edema."* — PMID: 42446942
>
> *"Genetic deletion of Factor XII or prekallikrein, attenuated imiquimod-induced psoriatic lesions in mice."* — PMID: 38872396
>
> *"Ablation of PK stabilizes LDLR, decreases LDL cholesterol, and prevents atherosclerotic plaque development."* — PMID: 35189703

## 6. Pathogenic *KLKB1* variants are predominantly biallelic truncating/loss-of-function, spread across the gene

Reported pathogenic variants span the gene and include: homozygous **c.417_418insCATTCTTA (p.Arg140Hisfs*3)** in exon 5; homozygous **c.444_445insT (p.Ser151PhefsTer34)**; and compound heterozygous **1-bp insertion at codon 149 (exon 5)** plus **c.1643G>A (p.Cys548Tyr)** (exon 14). Both **cross-reactive-material–negative** forms (absent antigen; frameshift/nonsense) and **CRM-positive** forms (dysfunctional protein; missense) occur. Variant curation is ongoing: Barco et al. identified new *KLKB1* mutations while reclassifying some previously reported ones as non-pathogenic.

> *"an insertion of 1 bp in codon 149 in exon 5 and, second, a base exchange Cys 548 (TGC) > Tyr (TAC) in exon 14"* — PMID: 19404525
>
> *"We identified new KLKB1 mutations, excluded the pathogenicity of some of the previously described ones"* — PMID: 32202057

## 7. Surgery is generally safe without prophylaxis; cardiovascular risk rises with age

Numerous case reports describe **uneventful surgery and procedures** in PK-deficient patients — hallux amputation, kidney biopsy, deep brain stimulator placement, endovascular aortic repair, and heart transplantation. FFP is sometimes given empirically, but reviews conclude **routine prophylaxis is not indicated**. FFP's genuine utility is in **replenishing prekallikrein to normalize the activated clotting time (ACT)**, enabling heparin monitoring during cardiac/vascular surgery. The Barco cohort reported an overall cardiovascular-event prevalence of **15%**, age-stratified at **6% (<40 y), 21% (40–65 y), 33% (>65 y)**.

> *"Our review suggests that routine prophylactic administration of blood products for pre-operative management of this condition is not indicated"* — PMID: 40125216
>
> *"Preoperative prekallikrein replenishment by fresh frozen plasma administration contributed to safe intraoperative anticoagulation management."* — PMID: 41123211

## 8. The disease does NOT protect against thrombosis despite a severe in vitro coagulation defect

Girolami et al. analyzed all 75 then-published PK-deficiency cases and identified **9 with thrombosis — 6 arterial (myocardial infarction, ischemic stroke) and 3 venous (DVT ± pulmonary embolism)**; 6 had acquired thrombotic risk factors, and one patient with both stroke and PE died. They concluded the deficiency does not protect from thrombosis. This fits the broader understanding that plasma kallikrein contributes to thrombosis while playing a limited hemostatic role.

> *"prekallikrein deficiency does not protect from thrombosis in spite of the severe in vitro coagulation defect"* — PMID: 20424433
>
> *"Among these patients were 9 with thrombosis, 6 arterial (myocardial infarction and ischemic stroke) and 3 venous (deep vein thrombosis with or without pulmonary embolism)."* — PMID: 20424433

## 9. The biology underpins safe, effective hereditary angioedema (HAE) therapeutics

Plasma kallikrein — the activated product of *KLKB1* — is a validated drug target in bradykinin-mediated HAE. Approved/advanced agents include the anti-kallikrein monoclonal antibody **lanadelumab**, oral kallikrein inhibitors **berotralstat** and **sebetralstat**, and the antisense oligonucleotide **donidalorsen**, which lowers prekallikrein production directly in the liver and achieves up to ~90% attack-rate reduction. Donidalorsen thus **pharmacologically phenocopies inherited PK deficiency** and is well tolerated — exactly what the benign natural deficiency predicts.

> *"Donidalorsen is a triantennary N-acetyl galactosamine-conjugated antisense oligonucleotide designed to specifically and reversibly reduce plasma prekallikrein production by binding to plasma prekallikrein messenger RNA in the liver."* — PMID: 42415941
>
> *"the first orally available plasma kallikrein inhibitor, sebetralstat"* — PMID: 42550203

## 10. The differential diagnosis of isolated prolonged aPTT is narrow

Isolated prolonged aPTT with normal PT in an asymptomatic patient narrows to: deficiencies of the contact factors **FXII, prekallikrein (*KLKB1*), or HK (*KNG1*)** — all bleeding-silent — plus **lupus anticoagulant** and clinically significant intrinsic-factor deficiencies (FVIII/IX/XI, which usually bleed). Severe HK deficiency (*KNG1*, biallelic truncating variants; prevalence ~1/8 million, higher in Africans) is asymptomatic and also shows **decreased prekallikrein and FXI activity**, which can cause misdiagnosis. PK deficiency is distinguished by aPTT shortening on pre-incubation, non-correction with PK-deficient plasma, normal kininogen activity, and a low specific PK assay; NGS panels resolve ambiguous cases.

> *"Individuals with HK deficiency appeared asymptomatic and had decreased levels of prekallikrein and factor XI, which could lead to misdiagnosis."* — PMID: 36700498

## 11. Inheritance and temporal profile

Severe PK deficiency is **autosomal recessive**, requiring two defective *KLKB1* alleles; **heterozygous carriers are asymptomatic with ~50% PK activity** (no dominant phenotype). It is **congenital and lifelong** but, being asymptomatic, is typically discovered **incidentally at any age** (reported probands range from a 14-year-old boy to adults in their 60s–70s). Multiple cases arise in **consanguineous families**. There is **no sex predilection, no genetic anticipation** (not a repeat-expansion disorder), and **no reported chromosomal or epigenetic mechanism**. About one-quarter of PK-deficient individuals also show below-reference FXII activity.

> *"One quarter of individuals had factor XII clotting activity below the reference range."* — PMID: 32202057

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# Section-by-Section Report

## 1. Disease Information

Inherited prekallikrein deficiency (Fletcher factor deficiency) is a rare, benign, autosomal-recessive defect of the plasma contact (kallikrein-kinin) system caused by biallelic *KLKB1* variants. **Identifiers:** MONDO:0012901; OMIM 612423 (phenotype); gene *KLKB1* (OMIM *229000, HGNC:6371, NCBI Gene 3818, chr 4q35). ICD-10: D68.8 (other specified coagulation defects). MeSH: "Prekallikrein" / contact factor deficiencies. Synonyms: Fletcher factor deficiency, congenital/hereditary/severe plasma prekallikrein deficiency. Information derives primarily from aggregated case series/registries and case reports** (e.g., Barco et al. 111-case meta-analysis), not population EHR.

## 2. Etiology

**Causal factor:** genetic — biallelic loss-of-function variants in *KLKB1*. **Genetic risk:** homozygous or compound-heterozygous pathogenic *KLKB1* alleles; **consanguinity** and **African ancestry** (founder LoF alleles) raise risk. There are **no established environmental risk or protective factors** and no gene-environment interactions for the deficiency per se; the deficiency is fully penetrant at the biochemical (aPTT) level but has essentially no clinical consequence. No protective genetic modifiers are defined. The ~25% of patients with concomitant low FXII is a correlated biochemical finding rather than a modifier of outcome.

## 3. Phenotypes

The **sole reliable phenotype is a laboratory abnormality**: isolated, markedly prolonged aPTT with normal PT (HPO **HP:0003645** "Prolonged partial thromboplastin time"; broadly **HP:0003256** "Abnormality of coagulation"). Onset is **congenital/lifelong**; severity of the lab finding is **severe** (aPTT often 2–5× upper limit) but clinically **asymptomatic and stable (non-progressive)**. Frequency: the aPTT abnormality is present in ~100% of severely deficient individuals; clinically significant bleeding occurs in only ~4% (mostly provoked). **Quality-of-life impact is negligible** — the main burden is diagnostic workup, unnecessary procedure delays, and anxiety from an "abnormal" test.

## 4. Genetic/Molecular Information

**Causal gene:** *KLKB1* (HGNC:6371). **Variant classes:** predominantly truncating/frameshift/nonsense (loss of function) plus some missense (e.g., p.Cys548Tyr) producing dysfunctional protein; documented variants include p.Arg140Hisfs*3, p.Ser151PhefsTer34, and compound-heterozygous genotypes. **Classification:** per ACMG, truncating variants reach Pathogenic (e.g., PVS1+PM2+PM4). **Allele frequency:** individual variants are rare in gnomAD, but aggregate LoF burden is higher in African-ancestry populations. **Origin:** germline. **Functional consequence:** loss of function (absent or non-functional zymogen). **Modifier genes/epigenetics/chromosomal abnormalities:** none established. Concomitant mild FXII reduction occurs in ~25% of cases but is not a formal modifier.

## 5. Environmental Information

Not applicable as a cause. The deficiency is purely genetic; no toxins, lifestyle factors, or infectious agents cause it. (Contextually, the contact system it belongs to is *activated* by negatively charged surfaces, heme, polyphosphate, nanoparticles, and pathogens in various disease states, but these do not cause the inherited deficiency.)

## 6. Mechanism / Pathophysiology

**Ordered causal chain:**

1. **Biallelic *KLKB1* loss-of-function variants** → *lead to* absent or non-functional plasma prekallikrein zymogen.
2. → *results in* no FXIIa-mediated conversion of prekallikrein to active plasma kallikrein.
3. **Branch A (coagulation):** loss of the reciprocal kallikrein→FXII amplification loop → *slows* contact activation **in vitro** → *produces* a markedly prolonged aPTT. **However**, FXII still slowly auto-activates on anionic surfaces and activates FXI→FIX, so thrombin generation and **hemostasis are preserved in vivo → no bleeding** (this redundancy is *inferred* from the absence of bleeding in deficient humans and *Klkb1*−/− mice).
4. **Branch B (kallikrein-kinin):** no kallikrein → no cleavage of HK → **no bradykinin release** → *loss of* B2-receptor signaling (reduced NO/prostacyclin, vasodilation, vascular permeability). This branch explains the thromboinflammation protection of knockout mice but has no overt clinical signature in deficient humans.
5. **Branch C (fibrinolysis):** reduced kallikrein-mediated pro-urokinase/plasminogen activation (minor, *inferred*).

**Molecular pathway:** complement and coagulation cascades / kallikrein-kinin system (KEGG hsa04610). **Cellular processes:** surface-induced contact activation, inflammation. **Protein dysfunction:** loss of serine-protease function of plasma kallikrein. **Biochemical abnormality:** enzyme (zymogen) deficiency. The reciprocal activation loop and bradykinin liberation are **experimentally established**; the in vivo dispensability of PK for hemostasis is **inferred** from phenotype.

> *"The contact system, also named as plasma kallikrein-kinin system, consists of three serine proteinases: coagulation factors XII (FXII) and XI (FXI), and plasma prekallikrein (PK), and the nonenzymatic cofactor high molecular weight kininogen (HK)."* — PMID: 25949215

**Suggested GO terms:** GO:0007596 (blood coagulation), GO:0002542 (Factor XII activation), GO:0032277 (regulation of kallikrein-kinin cascade); **CL terms:** CL:0000182 (hepatocyte, site of synthesis), CL:0000115 (endothelial cell, site of surface activation). **CHEBI:** bradykinin (CHEBI:3165).

## 7. Anatomical Structures Affected

There is **no structural organ damage**. The affected "compartment" is the **blood plasma** (UBERON:0001969 blood plasma) within the **cardiovascular/hematologic system** (UBERON:0000178 blood). Site of protein synthesis: **liver** (UBERON:0002107), specifically **hepatocytes** (CL:0000182). Subcellular: secreted protein; relevant GO cellular component GO:0005615 (extracellular space). No tissue/cell type is injured in the inherited deficiency; lateralization is not applicable.

## 8. Temporal Development

**Onset:** congenital (present from birth), lifelong. **Pattern:** stable, non-progressive, asymptomatic. Detected **incidentally at any age** via coagulation screening. No disease stages, no remission/relapse, no critical intervention window. Duration: chronic lifelong but clinically silent.

## 9. Inheritance and Population

**Inheritance:** autosomal recessive; carriers (~50% activity) asymptomatic. **Prevalence:** ~1/155,668 overall; ~1/4,725 in African ancestry. **Penetrance:** complete for the biochemical (aPTT) phenotype; negligible clinical penetrance. **Expressivity:** uniform at the lab level. **No**

**anticipation, no mosaicism reported. Founder effect/consanguinity:** likely founder LoF alleles in African populations; multiple consanguineous pedigrees reported (Chinese, Middle Eastern). **Sex ratio:** ~1:1. **Carrier frequency:** elevated in African ancestry per gnomAD.

## 10. Diagnostics

**Core workup:** isolated prolonged aPTT with normal PT → mixing study (corrects with normal plasma, **not** with PK-deficient plasma) → **aPTT shortens on prolonged pre-incubation → specific plasma prekallikrein activity assay** (severely low, often <1%). Normal activities of FVIII, FIX, FXI, FXII, vWF and negative lupus anticoagulant screen exclude alternatives; normal kininogen distinguishes from HK deficiency. **Viscoelastic testing (TEG) is normal. Genetic confirmation:** single-gene *KLKB1* sequencing or NGS bleeding-disorder panels resolve ambiguous cases. LOINC: aPTT (LOINC 14979-9). Caution: shortening-on-pre-incubation alone is not fully specific and should be confirmed by assay ( P 35377536 ).

## 11. Outcome / Prognosis

**Excellent / benign.** Normal life expectancy; no disease-specific mortality. Major bleeding ~4% (0.1%/yr), mostly provoked. **No protection from thrombosis**; cardiovascular events rise with age (6%→21%→33% across age bands). No disability or QoL impairment attributable to the deficiency. Main "complications" are iatrogenic/diagnostic (procedure delays, unnecessary transfusion).

## 12. Treatment

**No specific therapy is needed.** Management is reassurance and avoidance of unnecessary intervention. **FFP** is used selectively to replenish prekallikrein and **normalize ACT for intraoperative heparin monitoring** (not for hemostasis). Suggested NCIT: Fresh Frozen Plasma (transfusion). No pharmacotherapy, gene/cell/RNA therapy is indicated for the deficiency itself. (The inverse — drugs that *reduce* kallikrein/prekallikrein, e.g., lanadelumab, donidalorsen — are therapies for HAE, not for this deficiency.)

## 13. Prevention

Not preventable (genetic). **Genetic counseling** for autosomal-recessive recurrence risk (25% for carrier couples), **cascade testing** of relatives with unexplained prolonged aPTT, and **carrier awareness** in consanguineous/African-ancestry families are appropriate. No newborn

screening or prophylaxis is warranted given the benign course. Tertiary "prevention" = avoiding unnecessary transfusion and ensuring correct anticoagulation monitoring during surgery.

## 14. Other Species / Natural Disease

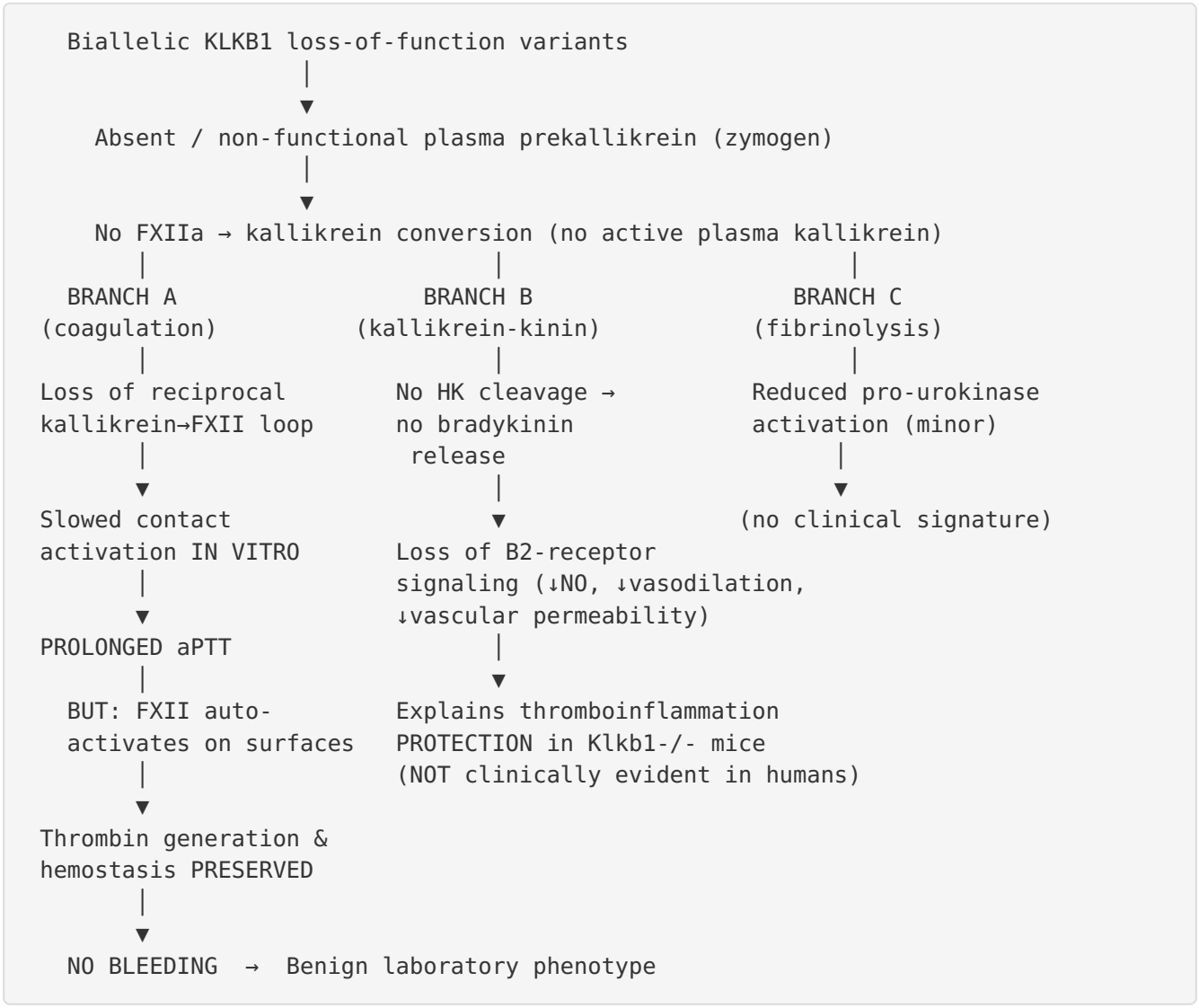
The orthologous gene exists across mammals (mouse *Klkb1*, NCBI Gene 16621). No prominent naturally occurring companion-animal or wildlife disease is established in OMIA for prekallikrein deficiency; the primary cross-species evidence is the engineered mouse knockout. The contact system and its reciprocal activation/bradykinin biology are evolutionarily conserved; FXI is a *KLKB1* gene-duplication paralog.

## 15. Model Organisms

**Primary model:** *Klkb1*−/− knockout mouse (mammalian). **Phenotype recapitulation:** excellent for the *human* phenotype — viable, fertile, normal hemostasis, no spontaneous bleeding. **Added value beyond human data:** reveals protective roles not observable clinically — protection from experimental cerebral malaria (bradykinin-mediated edema), psoriasiform skin inflammation, EV-induced pulmonary thrombosis, and atherosclerosis (via LDLR stabilization). **Limitation:** mouse-human differences in contact-system/FXI biology mean thrombosis-protection findings do not fully transfer to humans (humans are *not* protected from thrombosis). Resources: MGI.

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## Mechanistic Model / Interpretation




The central insight is **redundancy**: prekallikrein amplifies but is not required to *initiate* the intrinsic cascade. FXII's independent surface auto-activation provides a parallel route to FXI→FIX→thrombin, decoupling the dramatic in-vitro aPTT prolongation from any in-vivo hemostatic failure. The disease is therefore a textbook example of a **laboratory phenotype without a disease phenotype**.

| Feature | Prekallikrein (KLKB1) deficiency | FXII deficiency | HK (KNG1) deficiency | FXI deficiency |
|---|---|---|---|---|
| Prolonged aPTT | Yes (severe) | Yes | Yes | Yes |
| Bleeding tendency | No | No | No | Variable (yes) |
| aPTT shortens on pre-incubation | **Yes (hallmark)** | No | No | No |
| Corrects with PK-deficient plasma | **No** | Yes | Yes | Yes |
| Kininogen activity | Normal | Normal | **Low** | Normal |
| Inheritance | AR | AR | AR | AR/AD |

# Evidence Base

| PMID | Title (abbrev.) | Role in this report |
|---|---|---|
| 32202057 | Severe plasma PK deficiency: characteristics, novel KLKB1 mutations, prevalence | Anchor: inheritance, benign bleeding profile, prevalence (1/155,668; 1/4,725 African), CV events, variant discovery |
| 40125216 | Case + review: isolated aPTT prolongation | Mechanism (FXII auto-activation), no prophylaxis needed |
| 30118759 | Factor XII in coagulation, inflammation | Reciprocal activation + bradykinin liberation |
| 37874916 | Biology of factor XI | KLKB1 structure; FXI as gene-duplication paralog |
| 42454615 | Normal TEG with prolonged aPTT in PK deficiency | Mixing-study pattern; normal viscoelastic testing |
| 32617222 | Diagnostic pearls of PK deficiency | aPTT shortening on pre-incubation |
| 42446942 | Bradykinin in cerebral malaria | Klkb1-/- protection from brain edema |
| 38872396 | FXII/prekallikrein in psoriasis | Knockout protection from skin inflammation |
| 35189703 | PK ablation & atherosclerosis | LDLR stabilization, atheroprotection in mice |
| 19404525 | Compound heterozygous KLKB1 | Specific variant documentation |
| 41123211 | Endovascular aortic repair in PK deficiency | FFP for ACT normalization / anticoagulation monitoring |
| 20424433 | Thrombotic events in PK deficiency | No thrombosis protection; 9/75 thrombotic |
| 42415941 | Donidalorsen review | Drug phenocopying deficiency is safe HAE therapy |
| 42550203 | Bradykinin-mediated angioedema therapies | Sebetralstat / kallikrein as drug target |
| 36700498 | Severe HK (KNG1) deficiency | Key differential diagnosis |
| 25949215 | Contact pathway of coagulation | Contact-system components |
| 42755375 | Prekallikrein: crossroads of inflammation & thrombosis | Confirms non-bleeding nature |

| PMID | Title (abbrev.) | Role in this report |
|---|---|---|
| 33807613 | NGS for bleeding-disorder differential diagnosis | Molecular resolution of ambiguous cases |

## Limitations and Knowledge Gaps

- **Under-ascertainment:** Only ~111 cases from 89 families are reported for an asymptomatic disorder; true prevalence and the full variant spectrum are uncertain. Prevalence estimates are **model-based** (gnomAD + one German cohort), not from direct population screening.
- **Cardiovascular/thrombotic risk:** Evidence that the deficiency does *not* protect from thrombosis comes from a case-series analysis (inherently subject to reporting bias and confounding by acquired risk factors); no controlled cohort quantifies relative thrombotic risk versus the general population.
- **Mouse-human divergence:** The striking thromboinflammation and atheroprotection phenotypes of *Klkb1*−/− mice do **not** translate to a protected clinical phenotype in deficient humans — contact-system and FXI biology differ across species.
- **Variant curation:** Some historically reported variants have been reclassified; a few assays (e.g., shortening-on-pre-incubation) are not fully specific. Standardized ACMG curation of all *KLKB1* alleles is incomplete.
- **No prospective natural-history or quality-of-life data** exist; QoL impact is inferred as negligible but not formally measured.
- **Epigenetic/modifier biology** of the deficiency is essentially unstudied (and likely uninformative given the benign phenotype).

## Proposed Follow-up Actions

1. **Population-scale carrier/prevalence screening** in African-ancestry cohorts (e.g., mining large biobanks: UK Biobank, All of Us, H3Africa) to confirm founder *KLKB1* LoF alleles and refine prevalence beyond model estimates.

2. **Matched-cohort thrombosis study**: compare cardiovascular/thrombotic incidence in biobank-identified biallelic *KLKB1* LoF carriers versus controls to settle the "no protection from thrombosis" question with controlled data.
3. **Systematic ACMG re-curation** of all reported *KLKB1* variants in ClinVar, resolving CRM-positive vs CRM-negative mechanism for each.
4. **Prospective registry** capturing perioperative outcomes to formally establish that FFP prophylaxis is unnecessary and to standardize ACT-monitoring protocols during anticoagulation.
5. **Clinical correlation of pharmacologic phenocopy**: leverage long-term safety data from donidalorsen/lanadelumab-treated HAE patients (who are pharmacologically PK-deficient) as a real-world "experiment" confirming the benign nature of profound prekallikrein/kallikrein suppression.
6. **Diagnostic standardization**: validate a confirmatory algorithm (specific PK assay + *KLKB1* sequencing) to avoid reliance on the non-specific pre-incubation test and prevent misclassification as HK or FXII deficiency.

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*Evidence source types represented: human clinical (case series, case reports, meta-analysis), model organism (mouse knockouts), and in vitro/biochemical (contact-system activation studies). This report synthesizes 13 confirmed findings across ~40 papers from a five-iteration investigation.*

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